6.1.3 Manipulating genomes

Genome sequencing gives information about the location of genes and provides evidence for the evolutionary links between organisms.

Genetic engineering involves the manipulation of naturally occurring processes and enzymes. The capacity to manipulate genes has many potential benefits, but the implications of genetic techniques are subject to much public debate

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	the principles of DNA sequencing and the development of new DNA sequencing techniques	To include the rapid advancements of the techniques used in sequencing, which have increased the speed of sequencing and allowed whole genome sequencing e.g. high-throughput sequencing. HSW7
(b)	**(i)** how gene sequencing has allowed for genome-wide comparisons between individuals and between species **(ii)** how gene sequencing has allowed for the sequences of amino acids in polypeptides to be predicted **(iii)** how gene sequencing has allowed for the development of synthetic biology	With reference to bioinformatics and computational biology and how these fields are contributing to biological research into genotype–phenotype relationships, epidemiology and searching for evolutionary relationships. **PAG10** HSW7, HSW9
(c)	the principles of DNA profiling and its uses	To include forensics and analysis of disease risk. HSW9
(d)	the principles of the polymerase chain reaction (PCR) and its application in DNA analysis	
(e)	the principles and uses of electrophoresis for separating nucleic acid fragments or proteins	Opportunity for practical use of electrophoresis. **PAG6** HSW4
(f)	**(i)** the principles of genetic engineering	To include the isolation of genes from one organism and the placing of these genes into another organism using suitable vectors.
	(ii) the techniques used in genetic engineering	To include the use of restriction enzymes, plasmids and DNA ligase to form recombinant DNA with the desired gene and electroporation. HSW2

(g)	the ethical issues (both positive and negative) relating to the genetic manipulation of animals (including humans), plants and microorganisms	To include insect resistance in genetically modified soya, genetically modified pathogens for research and 'pharming' i.e. genetically modified animals to produce pharmaceuticals **AND** issues relating to patenting and technology transfer e.g. making genetically modified seed available to poor farmers. HSW10
(h)	the principles of, and potential for, gene therapy in medicine.	To include the differences between somatic cell gene therapy and germ line cell gene therapy. HSW9, HSW12